

VIBEGUARD: COLLEGE FABLAB 3D PRINTING & SLICER GUIDE

Fast Execution Card for Hardware Lead (Amith) & FabLab 3D Printer Technician

Target Rig: 10 Hz / 600 RPM Unbalance Testbed | Baseplate: Drilled Acrylic Stack | Total Est. Mass: ~10g PLA (Rs. 40–50) | Print Time: ~40 min

CRITICAL TECHNICIAN INSTRUCTIONS (DO NOT SKIP):

- 1. ZERO SUPPORTS (0%):** All parts are 100% self-supporting. Do NOT add support material inside the D-bore or wire window.
- 2. WALL PERIMETERS = 4 WALLS:** Ensures solid plastic around shaft bore and bolt recesses.
- 3. PRINT ORIENTATION:** Print all parts flat on their bottom faces (Z=0). Never print on edge.

1. QUICK SLICER SETTING MATRIX

Parameter	FAB-01: Rotor Cam Arm VibeGuard_Rotor_Arm_D_Shaft.stl	FAB-02: ADXL345 Mount VibeGuard_ADXL345_Rigid_Mount.stl	FAB-03: Motor Shims VibeGuard_N20_Motor_Shim_*.stl
Function	Couples to N20 D-shaft; spins at 600 RPM with M3 unbalance mass	Rigid accelerometer stand; rear window passes DuPont jumper wires	Eliminates 1.0–1.2 mm gap under motor cradle for zero vibration damping
Print Orientation	Flat on bottom face (Z=0)	Flat on base foot (Z=0)	Flat on build plate
Support Material	NONE (0% supports)	NONE (0% supports — 1mm top arch bridges cleanly)	NONE (0% supports)
Infill Density	100% Solid (Centrifugal stability)	80% to 100% (Rigid acoustic coupling)	100% Solid (High compressive clamp)
Perimeter Walls	Minimum 4 walls (>= 1.6 mm)	Minimum 4 walls (>= 1.6 mm)	3 walls
Layer Height	0.16 mm (Bore accuracy)	0.16 mm – 0.20 mm	0.16 mm – 0.20 mm
Est. Time & PLA	~12 to 14 min 3.5 g PLA	~20 to 22 min 5.5 g PLA	~3 to 4 min 0.8 g PLA

2. KEY FABRICATION & HARDWARE COMPATIBILITY FEATURES

- FAB-01 Rotor Arm (3.35mm & 3.40mm Failsafe):** Standard model features a 3.35 mm D-bore (chord 2.775 mm) for a snug push-fit. A second failsafe model (VibeGuard_Rotor_Arm_D_Shaft_3p40mm.stl) with 3.40 mm bore (chord 2.825 mm) is provided to guarantee zero jamming in case of printer over-extrusion. Both feature 12.0 mm eccentric pitch and 3.20 mm M3 bolt hole.
- FAB-02 Sensor Stand with Enlarged Window:** Upright M3 holes spaced 15.00 mm pitch at Z=7.0 mm matching ADXL345 PCB. Enlarged 21.6 mm (W) x 10.5 mm (H) rectangular window (Z=10.5 to 21.0 mm) fully clears the 20.32 mm black header base and 8 square DuPont female connectors (20.8 mm span). 1.0 mm top tie-bar bridges cleanly. (OpenTop U-frame variant also included in zip).
- FAB-03 Shim Options:** 3 options provided (1.0 mm flat, 1.2 mm flat, 1.0 mm cradle with side rails). Amith should print the 1.0 mm cradle shim + 1.0 mm flat shim to ensure perfect motor leveling under the white U-bracket.

3. PRE-HANDOVER INSPECTION CHECKLIST (WHAT AMITH MUST CHECK BEFORE LEAVING LAB)

- [] **1. D-Shaft Push-Fit:** Slide FAB-01 onto N20 motor shaft. It should slide on firmly by hand without radial wobble or slipping.
- [] **2. ADXL345 Alignment:** Place physical ADXL345 board against FAB-02 upright wall. Dual M3 holes line up (15.0 mm pitch).
- [] **3. Wire Window Clearance:** Confirm rectangular window is wide open (21.6x10.5 mm) with no drooping filament strings.
- [] **4. Base Flatness:** Confirm base flanges of FAB-02 and FAB-03 sit flat on a table with zero warping.
- [] **5. Part Integrity:** 100% solid, monolithic pieces — zero layer separation or splitting along layers.

Files inside ZIP: VibeGuard_FabLab_3D_Print_Pack.zip | Team: VibeGuard Hardware Subsystem Baseline | Contact: Amith Krishna Das